High Performans Analogue
MODEL ANSWERS
1.

(a) (Theory)

The student is expected to explain in their own words the problems associated to both eliminating the image signal and selecting the channel. These impose opposing constraints in filters which are very difficult and/or impossible to achieve in practice. The constraints can be relaxed with a double superheterodyne implementation. [4]

(b) (Theory)

To eliminate harmonics that can cause spurious. [4]

(c) (Application of Theory)

9.5MHz [4]

(d) (Application of Theory)

This is a lowpass filter, with a cut-off frequency which is lower than the frequency of the oscillator. Hence it would attenuate the main oscillation harmonic, and is wrongly designed.

(e) (New derivation. Extension of theory in the notes)

F= F1+(F2-1)/G1+(F3-1)/G1G2+(F4-1)/G1G2G3 [4]

2.

(a) (Theory)

Any version of a differential pair with any current mirror, any biasing and any type of transistor will be correct. [2]

(b) (Application of theory)

The answer of this will depend on what the students have drawn. I am expecting to see equivalent current and noise sources at the input of each transistor, and if there are resistors one source of thermal noise per resistor. [2]

(c) (Application of theory)

The most important noise sources are the ones of the input transistors (unless degeneration is used). Anything else will be attenuated when referred back to the input. [2]

(d) (New theory)

Worse. The equivalent input noise doubles because of noise sources being uncorrelated. [2]

(e) (Theory)

Better. In an, ideally, fully matched differential configurations, even order harmonics cancel themselves out. [2]

(f) (New theory)

Post-fabrication mismatch. [2]

(g) (Theory) [2]

Iout= Vin/R, where R is the degeneration resistance.

(h) (Application of theory) [3]

A very large resistance.

The negative effect is the added thermal noise of the resistor (which grows proportionally to its value and gets referred back to the input with an approximate gain of 1).

(i) (Application of theory)

The equivalent input noise would be the sum of the input noise of the input transistors plus the thermal noise of the resistances (the exact expression will depend on whether the students choose bipolar or MOS transistors). [3]

3.

(a) (New theory)

No, it is not correct. The exact expression will depend on the operating region of the transistors. [4]

(b) (Application of theory)

The transistors have to be properly sized. In order for the third harmonic to cancel

$$\frac{(W1/L1)^3}{(W2/L2)^3} = \frac{Is1}{Is2}$$ [4]

(c) (Application of theory)

The input voltage of the amplifier has not been chosen properly and M1 is not deep in strong inversion ohmic region. [4]

(d) (New theory)

One way is to make it differential. A second way is to choose a different value of Vb. [4]

(e) (New theory)

The feedback is making it oscillate (the amplifier's bandwidth is not high enough) [4]

4.

(a) (New theory)

The current is $I_{out} = \beta \left(\sum_{i=1}^{N} \frac{C_i}{C_{Total}} V_i - Vth \right)^2$ [4]

(b) (Application of theory)

Thermal and Flicker.

$ind^2 = \frac{8kTgm\Delta f}{3}\ A^2$

and $vng^2 = \frac{k_f}{CoxWLf}\ V^2/Hz$ [4]

(c) (New theory)

Worse, since the noise will be amplified by power gain from the floating gate to the equivalent input, which is larger than one (since that voltage is attenuated from the input to the gate by the ratio between the input capacitance and the total capacitance seen by the gate). [2]

(d) (New theory)

It is the same. The noise sources referred to the gate are the same and there is no variation of gain from the gate to the output. [2]

(e) (New theory)

The transistor is more linear for larger inputs, because the voltage seen by the gate is attenuated by the ratio between the input capacitance and the total capacitance. [2]

(f) (New theory)

Open question. The explanation of the choices will be evaluated, regardless of the circuit invented. [6]